

Keith J. Polson
Senior Vice President and
Chief Nuclear Officer

DTE Energy Company
6400 N. Dixie Highway, Newport, MI 48166
Tel: 734.586.6515 Fax: 734.586.1431
Email: keith.polson@dteenergy.com



10 CFR 50.73

September 29, 2018
NRC-18-0047

U. S. Nuclear Regulatory Commission
Attention: Document Control Desk
Washington, D.C. 20555-0001

Reference: Fermi 2
NRC Docket No. 50-341
NRC License No. NPF-43

Subject: Licensee Event Report (LER) No. 2018-005

Pursuant to 10 CFR 50.73(a)(2)(v)(D), DTE Electric Company (DTE) is submitting LER No. 2018-005, Non-Functional Mechanical Draft Cooling Tower Fan Brakes Leads to HPCI Being Declared Inoperable and Loss of Safety Function.

No new commitments are being made in this LER.

Should you have any questions or require additional information, please contact Mr. Scott A. Maglio, Manager – Nuclear Licensing, at (734) 586-5076.

Sincerely,

Keith J. Polson
Senior Vice President and CNO

Enclosure: Licensee Event Report No. 2018-005

cc: NRC Project Manager
NRC Resident Office
Reactor Projects Chief, Branch 5, Region III
Regional Administrator, Region III
Michigan Public Service Commission
Regulated Energy Division (kindschl@michigan.gov)

**Enclosure to
NRC-18-0047**

**Fermi 2 NRC Docket No. 50-341
Operating License No. NPF-43**

**Licensee Event Report (LER) No. 2018-005
Non-Functional Mechanical Draft Cooling Tower Fan Brakes Leads to HPCI Being
Declared Inoperable and Loss of Safety Function**

NRC FORM 366 (04-2017)		U.S. NUCLEAR REGULATORY COMMISSION			APPROVED BY OMB: NO. 3150-0104		EXPIRES: 03/31/2020				
LICENSEE EVENT REPORT (LER) (See Page 2 for required number of digits/characters for each block) (See NUREG-1022, R.3 for instruction and guidance for completing this form http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)											
1. FACILITY NAME Fermi 2					2. DOCKET NUMBER 05000 341		3. PAGE 1 OF 4				
4. TITLE Non-Functional Mechanical Draft Cooling Tower Fan Brakes Leads to HPCI Being Declared Inoperable and Loss of Safety Function											
5. EVENT DATE			6. LER NUMBER			7. REPORT DATE			8. OTHER FACILITIES INVOLVED		
MONTH	DAY	YEAR	YEAR	SEQUENTIAL NUMBER	REV NO.	MONTH	DAY	YEAR	FACILITY NAME	DOCKET NUMBER	
08	03	2018	2018	005	0	09	29	18	FACILITY NAME	DOCKET NUMBER	
										05000	
										05000	
9. OPERATING MODE		11. THIS REPORT IS SUBMITTED PURSUANT TO THE REQUIREMENTS OF 10 CFR §: (Check all that apply)									
1		☐ 20.2201(b)			☐ 20.2203(a)(3)(i)			☐ 50.73(a)(2)(ii)(A)		☐ 50.73(a)(2)(viii)(A)	
		☐ 20.2201(d)			☐ 20.2203(a)(3)(ii)			☐ 50.73(a)(2)(ii)(B)		☐ 50.73(a)(2)(viii)(B)	
		☐ 20.2203(a)(1)			☐ 20.2203(a)(4)			☐ 50.73(a)(2)(iii)		☐ 50.73(a)(2)(ix)(A)	
		☐ 20.2203(a)(2)(i)			☐ 50.36(c)(1)(i)(A)			☐ 50.73(a)(2)(iv)(A)		☐ 50.73(a)(2)(x)	
100		☐ 20.2203(a)(2)(ii)			☐ 50.36(c)(1)(ii)(A)			☐ 50.73(a)(2)(v)(A)		☐ 73.71(a)(4)	
		☐ 20.2203(a)(2)(iii)			☐ 50.36(c)(2)			☐ 50.73(a)(2)(v)(B)		☐ 73.71(a)(5)	
		☐ 20.2203(a)(2)(iv)			☐ 50.46(a)(3)(ii)			☐ 50.73(a)(2)(v)(C)		☐ 73.77(a)(1)	
		☐ 20.2203(a)(2)(v)			☐ 50.73(a)(2)(i)(A)			☒ 50.73(a)(2)(v)(D)		☐ 73.77(a)(2)(i)	
		☐ 20.2203(a)(2)(vi)			☐ 50.73(a)(2)(i)(B)			☐ 50.73(a)(2)(vii)		☐ 73.77(a)(2)(ii)	
					☐ 50.73(a)(2)(i)(C)			☐ OTHER		Specify in Abstract below or in NRC Form 366A	
12. LICENSEE CONTACT FOR THIS LER											
LICENSEE CONTACT Fermi 2 / Scott A. Maglio - Manager, Nuclear Licensing								TELEPHONE NUMBER (Include Area Code) 734-586-5076			
13. COMPLETE ONE LINE FOR EACH COMPONENT FAILURE DESCRIBED IN THIS REPORT											
CAUSE	SYSTEM	COMPONENT	MANU-FACTURER	REPORTABLE TO EPIX	CAUSE	SYSTEM	COMPONENT	MANU-FACTURER	REPORTABLE TO EPIX		
Unk	120VAC	FU	Fusetron	Yes	NA	NA	NA	NA	NA		
14. SUPPLEMENTAL REPORT EXPECTED ☐ YES (If yes, complete 15. EXPECTED SUBMISSION DATE) ☒ NO					15. EXPECTED SUBMISSION DATE						
					MONTH DAY YEAR						
ABSTRACT (Limit to 1400 spaces, i.e., approximately 15 single-spaced typewritten lines) At 0940 EDT on August 3, 2018, the Division 2 Mechanical Draft Cooling Tower (MDCT) fans were declared inoperable due to loss of power to the over speed fan brake inverter. The MDCT fans are required to support operability of the Ultimate Heat Sink (UHS) and the Emergency Equipment Cooling Water (EECW) system. The Division 2 EECW system cools the High Pressure Coolant Injection (HPCI) system room cooler. As a result, the non-functionality of the fan brakes led to an unplanned HPCI inoperability. Since HPCI is a single train system designed to mitigate the consequences of a loss of coolant accident (LOCA), this event could have prevented the fulfillment of a safety function. The cause of the event was the failure of the Division 2 fan brake inverter. Corrective Actions were taken to replace the blown fuse and inverter returning the MDCT fans, the UHS, EECW and HPCI to service on August 4, 2018 at 2151 EDT. The fan brake system is only required for a design basis tornado and there was no credible tornado threat during this event. The inverter was fully tested by Fermi 2 with assistance and guidance from the vendor NOVA. All testing was satisfactory. The fuse has been sent for failure analysis. The HPCI system is not required to mitigate a design basis tornado. The safety significance of this event is very low and there were no radiological releases associated with this event.											

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the Information Services Branch (T-2 F43), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOB-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
Fermi 2	05000-05000-341	YEAR 2018	SEQUENTIAL NUMBER 005	REV NO. 0

NARRATIVE**INITIAL PLANT CONDITIONS**

Mode – 1

Reactor Power – 100 percent

There were no structures, systems, or components (SSCs) that were inoperable at the start of this event that contributed to this event.

DESCRIPTION OF THE EVENT

At 0940 EDT on August 3, 2018, the Division 2 Mechanical Draft Cooling Tower (MDCT) [CTW] fans [FAN] were declared inoperable due to loss of power to the over speed fan brake [BRK] inverter [INVT]. The brakes prevent the fans from experiencing over speed from a design basis tornado as described in Updated Final Safety Analysis Report (UFSAR) Section 9.2.5.2.2. The MDCT fans are required to support operability of the Ultimate Heat Sink (UHS) [BS]. The UHS is required to support operability of the Division 2 Emergency Equipment Cooling Water (EECW) system. The EECW system cools various safety related components, including the High Pressure Coolant Injection (HPCI) [BJ] system room cooler [CLR]. As a result, the non-functionality of the fan brakes led to an unplanned HPCI inoperability based on the inoperability of the EECW to the HPCI room cooler. Following restoration of the brake inverter, HPCI was declared operable on August 4, at 2151 EDT. Therefore, the single train HPCI system was inoperable for approximately 36 hours and 11 minutes. This report is being made pursuant to 10CFR50.73(a)(2)(v)(D) as a condition that could have prevented the fulfillment of a safety function needed to mitigate the consequences of an accident.

Technical Specification 3.7.2 requires the UHS reservoir to be operable in Modes 1, 2, and 3. The UHS operability requirements in Modes 4 and 5 are determined by the systems the UHS supports. The UHS reservoir is divided into two one-half capacity reservoirs, corresponding to Division 1 and Division 2. Each reservoir is the cooling source for that division's service water subsystems, including the service water for the Emergency Diesel Generators (EDGs) [[DG]]. A two-cell MDCT is located above each of the one-half capacity reservoirs. Each cell is equipped with a MDCT fan. Two MDCT fans above each one-half capacity reservoir are required for it to be considered operable. The "A" and "C" MDCT fans are in Division 1 and the "B" and "D" fans are in Division 2. The MDCT fans have a brake system to prevent overspeed from a design basis tornado as described in UFSAR Section 9.2.5.2.2. Thus the fan brakes are a design feature of the fans and their functionality impacts the operability of the MDCT fans and therefore the UHS.

The brake mechanism senses the speed of the motor and blades and uses nitrogen pressure to activate a brake system to slow the fan speed. The control power for this braking control module is powered by the R1700S011B (Division 2 Residual Heat Removal (RHR) MDCT Fans overspeed Brake Inverter). Without this inverter being energized, neither of the Division 2 MDCT fan brakes would perform their design function. Therefore, this inverter failure renders both of the Division 2 MDCT fans inoperable.

A past operability review was performed and concluded that the MDCT Brake Inverter Panel R1700S011B was discovered to be indicating no AC Volts or Amps during daily rounds on August 3, 2018 at 0940. An Operator records the output voltage daily and it was observed that the output voltage was within acceptable limits on August 2, 2018 at 1035 EDT. Thus, the inverter was observed to be functioning 23 hours 5 minutes before discovery and the maximum possible period of inverter failure prior to discovery and restored to service is approximately 59 hours and 16 minutes.

A review verified that the Division 1 UHS and RCIC systems were operable during the possible period of inverter failure, as well as during the period when Division 2 UHS and HPCI were declared inoperable. Therefore, there was not a violation of Technical Specifications and Licensing Basis due to this condition. TS 3.7.2 Condition B allows for 72 hours

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1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
Fermi 2	05000- 341	YEAR 18	SEQUENTIAL NUMBER 005	REV NO. 00

NARRATIVE

of UHS unavailability and TS 3.5.1 Condition E allows for 14 days of HPCI unavailability as long as the Reactor Core Isolation Cooling (RCIC) system [BN] is operable.

Additionally, Note 1 of LCO 3.7.2 requires entering applicable Conditions and Required Actions of LCO 3.8.1 for EDGs. At the time when LCO 3.7.2 was entered (August 3, 2018 at 0940), LCO 3.8.1 was also entered. For the 23 hours and 5 minute period of potentially inoperability, LCO 3.8.1 was not entered; however, offsite power was available and all actions of LCO 3.8.1 would have been met.

SIGNIFICANT SAFETY CONSEQUENCES AND IMPLICATIONS

As described previously, the MDCT fan brakes are only required for a design basis tornado; they are not required for other design basis events. No tornado watches or warnings occurred during the time-frame when MDCT fan brakes were nonfunctional. Since there was no credible threat of a tornado the MDCT fans and associated UHS would have been capable of performing their intended functions throughout the period described above.

As discussed above the nonfunctionality of the Division 2 MDCT fan brakes could have prevented the HPCI system from fulfilling its safety function. However, the RCIC system remained available for high pressure injection in the event of an emergency. The Standby Feedwater [SJ] system was also available for high pressure injection. Additionally, the Automatic Depressurization System [JE] was available to reduce reactor pressure to within the capabilities of the low pressure Emergency Core Cooling Systems. HPCI could have been prevented from fulfilling its safety function as a result of the MDCT fan brake nonfunctionality for a possible maximum of 59 hours and 16 minutes. The previous reading of the inverter occurred August 2 at 1035 EDT 23 hours and 05 minutes prior to discovery and an additional 36 hours and 11 minutes following discovery to replace the inverter and restore the MDCT fan and HPCI to Operable. TS 3.5.1 allows HPCI to be taken out of service for planned outages for up to 14 days. It is important to note that, although HPCI was considered inoperable due to the propagation of TS requirements for UHS and was therefore considered a loss of safety function in this LER, HPCI is not a system required for safe shutdown in the event of a tornado per UFSAR Table 3.3-2. HPCI is required for design basis accidents such as a loss of coolant accident (LOCA). Therefore, although HPCI was declared inoperable at the time, HPCI would have been capable of performing its intended LOCA function regardless of the inverter failure. As described in Regulatory Guide 1.117, design basis tornados are not postulated to occur simultaneously with a LOCA. Based on this discussion, the safety significance of this event is very low. There were no radiological releases associated with this event.

CAUSE OF THE EVENT

The cause of this event is due to a failed fuse which caused a loss of output to the over speed fan brake inverter. The inverter was fully tested satisfactory.

Cause: Fuse Failure
System: R3200 – DC Systems
Component: Fuse
Manufacturer: Fusetron

CORRECTIVE ACTIONS

The fuse and inverter were replaced and the MDCT was returned to service on August 4, 2018 at 2151 EDT.

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NARRATIVE**PREVIOUS OCCURRENCES**

Previous occurrences of HPCI inoperability due to non-functionality of the MDCT fan brake system were reported in LERs 2016-006 and 2017-005. However, the instances described in LER 2016-006 were associated with inadequate procedural guidance related to the fan brake systems and were all the result of problems with the nitrogen pressure supply. The condition described in this LER is not a result of inadequate procedural guidance and was not due to problems with the nitrogen pressure supply. Therefore, the corrective actions previously performed in LER 2016-006 could not have prevented the occurrence described in this LER as they could not have prevented the inverter from failing.

The instance described in LER 2017-005, Non-Functional Mechanical Draft Cooling Tower Fan Brakes Leads to HPCI Being Declared Inoperable and Loss of Safety Function, was a result of the loss of inverter output. However, there was no loss of the input fuse. Therefore, the corrective actions previously performed in LER 2017-005 could not have prevented the occurrence described in this LER as they could not have prevented the input fuse from failing.